

QINGYU ZHANG (Alfred)

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Web: <https://alfredzhang98.github.io/> (More Project Details Here)



Who I am

Qingyu Zhang is a PhD researcher in Electronic and Electrical Engineering at University College London (UCL), jointly funded by EPSRC and the China Scholarship Council (CSC). He won the Best Live Demo Award at IEEE ISCAS 2025 for his work on haptic-enhanced bioimpedance needles. He graduated top of his MSc class at the University of Bristol (UoB) with distinction and previously worked in industry as an Internet of Things (IoT) engineer and project manager, co-inventing six patents. He was also awarded the Outstanding Employee of the Quarter in 2021 at New Oriental's AI research institute. His research interests span bioelectronic circuit and system design, medical AI and deep learning, reinforcement learning for robotics and haptics, and vision language action models.

EDUCATION

PhD in Electronic and Electrical Engineering, UCL	Mar 2023 – Now
MSc in Advanced Microelectronic System Engineering (AMSE), UoB	Sep 2022 - Sep 2023
- Degree Class: Distinction (1st place in the class of years 22-23, overall score 83/100) - Final project mark: 88/100 - Student representative of AMSE	
BEng in Optical Electronic Information Engineering, Huazhong University of Science and Technology	Sep 2016 - Jul 2020
- Overall grade: GPA 3.22/4 - Final project mark: 88/100	

WORK EXPERIENCE

IoT Manager, New Oriental Education Technology Group (XDF), Beijing, China	Feb 2021 - Dec 2021
- Led a team of three IoT engineer colleagues. - Developed a full-stack controller based on the ESP32 chip for smart classroom applications, featuring a “develop once, adapt everywhere” architecture. Integrated BLE, audio, IMU, and multiple peripheral button interfaces. - Over 200 controllers were produced when I left the company.	
IoT Engineer, Shensi Shudun Technology, Co, Ltd, Beijing, China	Jul 2020 - Jan 2021
- In charge of developing Windows desktop applications for streamlining the assembly production line based on PyQt. - Led the development of a smart dock that enhanced the functionality of existing air purifiers, based on STM32 and ESP8266 platforms, featuring one-click network configuration and multiple communication interfaces including Wi-Fi, UART, and I²C. - Re-structured the hardware communication process on the M263KI chip and successfully expanded SPI communication to 6 serial channels, significantly enhancing efficiency compared to the existing four-channel scheme. - The smart dock is mass-produced at 1000+ units per year.	

PUBLICATIONS

Conferences Paper:

- Jiayang Li, **Qingyu Zhang**, Dai Jiang, Sohmyung Ha, Andreas Demosthenous, and Yu Wu, "A 0.62 μ W/sensor 82 fps Time-to-Digital Impedance Measurement IC with Unified Excitation/Readout Front-end for Large-Scale Piezo-Resistive Sensor Array," submitted to IEEE International Solid-State Circuits Conference (ISSCC), 2026.
- Q. Zhang**, A. Al-Hindawi, A. Demosthenous, and Y. Wu, “Haptic enhanced bioimpedance needle for precision navigation in central venous catheterisation with millimetre accuracy,” submitted to *2025 IEEE Int. Symp. Circuits Syst.*
- J. Chi, **Q. Zhang**, Z. Zhang, A. Demosthenous, and Y. Wu “High Resolution Plantar Pressure Insole System for Enhanced Lower Body Biomechanical Analysis,” submitted to *2025 IEEE Int. Symp. Circuits Syst.*
- J. Zhang, **Q. Zhang**, Y. Wu, D. Jiang and A. Demosthenous “A Method for Improving the Wearability of Capsule Positioning System Using Particle Filter,” submitted to *2025 IEEE Int. Symp. Circuits Syst.*

Live Demo Paper:

- Q. Zhang**, A. Al-Hindawi, A. Demosthenous, and Y. Wu, “Live Demonstration: Haptic-Enhanced Bioimpedance Needle for Assisting Central Venous Catheterisation,” submitted to *2025 IEEE Int. Symp. Circuits Syst.*
- J. Chi, **Q. Zhang**, Z. Zhang, A. Demosthenous, and Y. Wu, “Live Demonstration: A High-Resolution Plantar Insole System for Lower Body Estimation,” submitted to *2025 IEEE Int. Symp. Circuits Syst.*

PATENTS

- ‘Image processing method and apparatus, electronic device and computer-readable storage medium’, CN115393922A, 2022
- ‘Data processing methods, apparatus, storage media and electronic devices’, CN114969790A, 2022
- ‘Attitude recognition method, apparatus, storage medium and electronic device’, CN114782994A, 2022
- ‘Sitting posture recognition method, device and storage medium’, CN114821652A, 2022
- ‘Face recognition method, apparatus, storage medium and electronic device’, CN114998955A, 2022
- ‘Image processing method, apparatus, storage medium and electronic device’, CN114913181A, 2022

AWARDS

- Best live demo, ISCAS, 2025
- Outstanding Employee of the Quarter in 2021, top 1.5% of 200 colleagues in the AI research institute of XDF
- Outstanding graduate from HUST in BEng and BA college in 2020
- Academic Progress Scholarship in HUST in 2019
- Outstanding Student Cadres Scholarship in HUST in 2018 and 2019
- Scientific and Technological Innovation Scholarship in HUST in 2018
- Third prize in the Mathematical Modelling Competition in central China in 2018, the top 5% of 2000 groups
- Second prize at HUST Entrepreneur Camp in 2018, second out of 26 teams

SKILLS

- **Coding Skills:** C/C++ (8/10), Python (7/10), Matlab (4/10), Verilog (5/10), Typescript (2/10)
- **Embedded Systems:** Proficient in development across Linux, embedded RTOS, and Android platforms
- **3D Design & Simulation:** Skilled in SolidWorks, Blender, and COMSOL for 3D modelling and simulation, with specialized experience in co-simulating electrical properties, particularly electric field analysis for bioimpedance
- **Circuit Design:** Experienced in PCB schematic design, layout optimisation, and hardware debugging across power management systems, embedded system integration, mixed-signal (analog/digital) circuits, and application-specific integrated circuits
- **Machine Learning Skills:** Pytorch (6/10), with experience in computer vision (e.g., segmentation, classification) and reinforcement learning
- **Robotic Platform:** Hands-on experience with UR collaborative robots, haptic devices (Haply Inverse3), and custom-built robotic prototypes
- **Research and Research Writing Skills**

RESEARCH & PROJECTS [\(Click Here To Read Details\)](#)

<i>Research assistant: Developing the smart insole related hardware and software system</i>	Mar 2025 – Sep 2025
<i>Research assistant: Developing the ultra-low power embedded system for in vivo stimulation</i>	Oct 2024 – Apr 2025
<i>PhD research: Bioimpedance needle trajectory system</i>	May 2024 - Oct 2024
• Performed bioimpedance needle structure analysis using COMSOL. • Developed a complete application system integrating an embedded Bluetooth module, a haptic robot, and a user interface. • Key result: In 24 user-participation trials on a phantom model, the system achieved a 100% success rate for venous entry, with a root mean square error of 0.66 mm and an 87% probability of positioning within ± 1 mm of the vein centre. This system has been evaluated by NHS colleagues for medical needle navigation.	
<i>MSc project: design and modelling of power conditioning circuits for triboelectric nanogenerator</i>	May 2023 - Sep 2023
• Used COMSOL for finite element analysis to model and analyse a triboelectric nanogenerator (TENG). • Used Verilog-AMS in Cadence to design an analytical TENG model. • Based on TENG models to design a reliable and efficient power management circuit for the bladder actuator. • Key result: the system can generate 3.25 mAh of electricity daily with an efficiency of 23%, which is higher than most designs.	
<i>BEng project: design and optimisation of geomagnetic algorithm based on DSP</i>	Feb 2020 - Jun 2020
• Optimise the World Magnetic Model 2020 (WMM2020) algorithm. • Implement this algorithm on a DSP platform and focus on hardware performance improvement. • Key result: achieved relative error of less than 0.003% and improved calculation speed to approximately 0.57ms, which is twice the performance of existing systems at the time.	
<i>Research Assistant at China National Photovoltaic Laboratory</i>	Oct 2019 - Jan 2020
• The project aims to measure blood flow rate and oxygen level by laser scattering imaging, funded by HUAWEI. • In charge of setting the camera and laser experimental platform. • In charge of optimising the laser scatter imaging algorithms. • Key results: generated a thermal map to indicate the organism's blood flow rate and a monitoring curve of blood oxygen over time. Compared to other methods, these results allow blood flow rate and oxygen level to be obtained by standard mobile phones.	